



Yesterday's Count, Tomorrow's Rain

Forecasting vs Naive Restocking of the Campus Loaner-Umbrella Bins

Fenna Okoro · Ingrid Vasseur — School of Emergent Priorities

Background

Wet mornings empty the campus umbrella bins within the hour; dry ones leave them stocked and unused. The Horizon Register forecasts wet-weather demand three days out. Does the forecast actually improve next-day restocking over a caretaker who simply repeats yesterday's count?

Aims

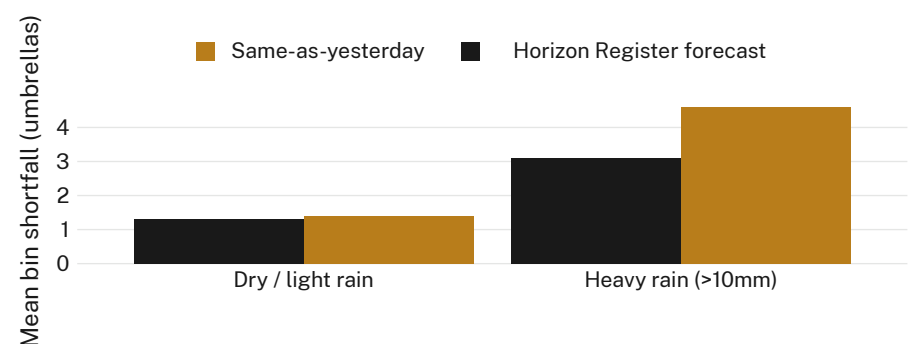
- Compare next-day restocking accuracy under the Register's forecast rule against a same-as-yesterday rule
- Establish whether any forecast advantage holds uniformly or only above a rainfall threshold
- Estimate the caretaking cost of switching rules

Methods

- 6 loaner-umbrella bins across two schools · daily stock counts · 14-week term
- Register forecast probability compared against a persistence baseline (yesterday's closing count)
- Rainfall bands pre-registered; each day's restocking decision logged before the count, never adjusted post hoc

Results

Restocking shortfall under the Register's forecast is indistinguishable from the baseline on dry and light-rain mornings. On days with forecast rainfall above 10mm, the forecast rule cuts mean shortfall by roughly a third.



Mean daily umbrella shortfall on the six monitored bins fell from 4.6 to 3.1 units on forecast-heavy-rain mornings between weeks 3 and 11 of Semester 1, 2026 (n = 84 bin-days); shortfall on drier mornings showed no reliable difference between rules.

Discussion & conclusions

The Register's forecast earns its keep only when it is least convenient to ignore: heavy-rain mornings. On ordinary days it adds no detectable value over a caretaker's memory of yesterday, and the saving surfaces only where the baseline was already failing hardest. Next: extending the audit to the remaining campus bins across a full academic year.

References

1. Murphy, A. H. (1992). Climatology, persistence, and their linear combination as standards of reference in skill scores. *Weather and Forecasting*, 7(4), 692–698. doi:10.1175/1520-0434(1992)007<0692:CPATLC>2.0.CO;2
2. Armstrong, J. S. (2006). Findings from evidence-based forecasting: Methods for reducing forecast error. *International Journal of Forecasting*, 22(3), 583–598. doi:10.1016/j.ijforecast.2006.04.001
3. Fildes, R., Goodwin, P., Lawrence, M., & Nikolopoulos, K. (2009). Effective forecasting and judgmental adjustments: an empirical evaluation and strategies for improvement in supply-chain planning. *International Journal of Forecasting*, 25(1), 3–23. doi:10.1016/j.ijforecast.2008.11.010
4. Taghizadeh, E. (2017). Utilizing artificial neural networks to predict demand for weather-sensitive products at retail stores. arXiv:1711.08325
5. Gammelli, D., Wang, Y., Prak, D., Rodrigues, F., Minner, S., & Pereira, F. C. (2021). Predictive and prescriptive performance of bike-sharing demand forecasts for inventory management. arXiv:2108.00858

Supported by a School of Emergent Priorities seed allocation; we thank the caretaking staff of the monitored entrances.

Rainfall bands pre-registered; bin counts logged by rostered caretaking staff; no umbrellas retained or identified.

Office of Research Outputs · slop.university